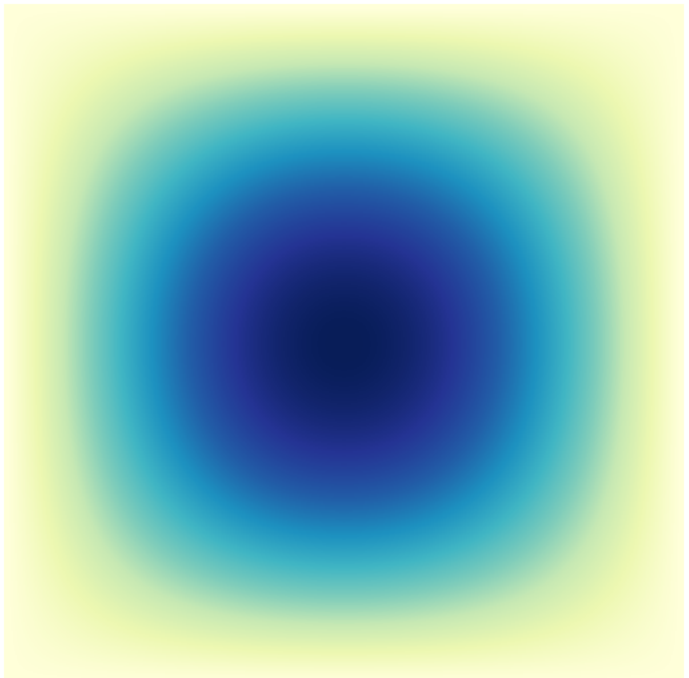
Variable diffusion

Conservative discretisation and numerical verification

Group D



Computed temperature · N = 64

Steady heat in heterogeneous materials

Layered composites, soil and tissue

$$-\nabla \cdot (a \nabla u) = f$$

u Temperature
a Thermal conductivity
f Internal heat source

$$u = g \text{ on } \partial\Omega, \qquad \Omega = (0,1)^2$$

$$0 < a_{\min} \leq a(x,y) \leq a_{\max}$$

$$\mathbf{q} = -a \nabla u$$

$$\nabla \cdot \mathbf{q} = f$$

Steady conservation with prescribed boundary temperature and ideal thermal contact

A smooth manufactured benchmark

Choose the exact temperature, then derive the source

$$a = 1 + x, \qquad u_* = \sin(\pi x)\sin(\pi y)$$

Manufactured forcing

$$f = 2\pi^2(1 + x)\sin(\pi x)\sin(\pi y) - \pi\cos(\pi x)\sin(\pi y)$$

$$\mathbf{q}_* = -\pi(1 + x)(\cos(\pi x)\sin(\pi y),\ \sin(\pi x)\cos(\pi y))$$

g = 0 on all sides. Temperature and heat flux are continuous. Symbolic differentiation checks f independently

A layered interface benchmark

aL = 1 aR = 10 f = 0 interface x = 1/2

$$u_*(x) = \frac{20x}{11} \quad \left(x \le \frac{1}{2}\right)$$

$$u_*(x) = \frac{2x+9}{11} \quad \left(x \ge \frac{1}{2}\right)$$

Ideal interface conditions

$$u_L\left(\tfrac{1}{2}\right) = u_R\left(\tfrac{1}{2}\right) = \frac{10}{11}$$

$$-a_L u_L' = -a_R u_R' = -\frac{20}{11}$$

Temperature and normal flux are continuous. The slope jumps.

Full analytic Dirichlet data on all sides. Odd N places the interface midway between nodal columns

Conservative five-point FDM

Nodal temperatures on a uniform grid, h = 1/N

$$q_E = -a_E \frac{U_{i+1,j} - U_{i,j}}{h}$$

$$(A_h U)_{i,j} = \frac{1}{h^2}\big[a_E(U_{i,j} - U_{i+1,j}) + a_W(U_{i,j} - U_{i-1,j}) + a_N(U_{i,j} - U_{i,j+1}) + a_S(U_{i,j} - U_{i,j-1})\big] = f_{i,j}$$

One centre and four neighbours. Each face coefficient is shared.

Flux balance gives the conservative five-point stencil and a sparse interior system AU = b

Arithmetic and harmonic averages

Smooth conductivity

$$a_{\text{arith}} = \frac{a_L + a_R}{2}$$

Arithmetic midpoint approximation
Exact midpoint for a = 1+x

Material interface

$$a_{\text{face}} = \frac{h}{d_L/a_L + d_R/a_R}$$

$$a_{\text{harm}} = \frac{2a_L a_R}{a_L + a_R}$$

For aL = 1 and aR = 10: arithmetic 5.5, harmonic 20/11

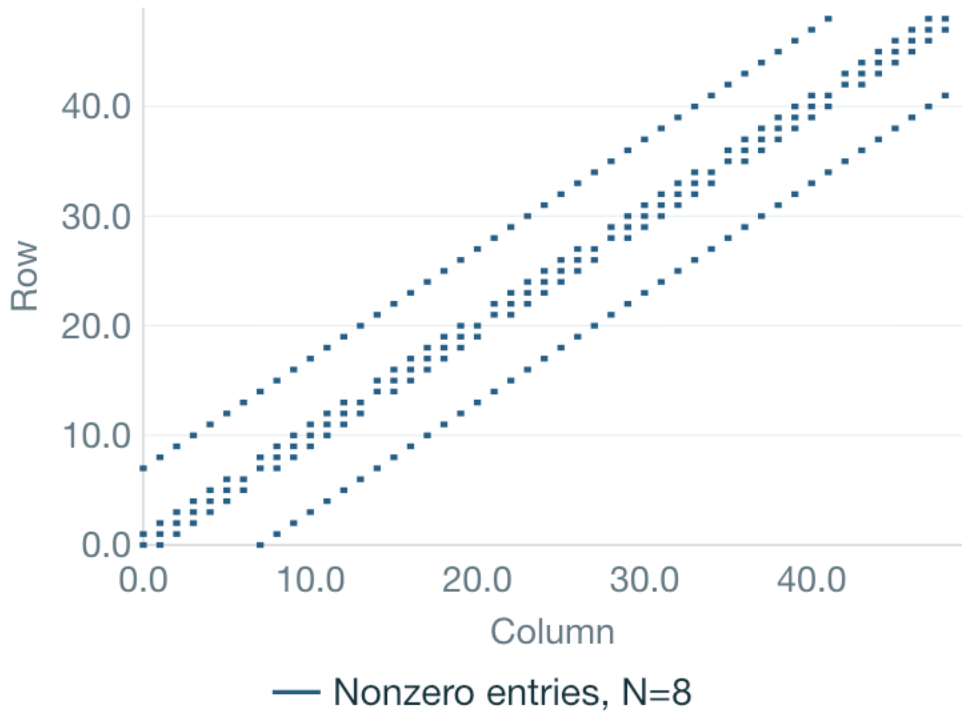
Thermal resistances add in series. The harmonic formula assumes equal material distances on the cut edge

Five tasks and their evidence

Task	Question	Evidence
1	Correct discrete matrix?	Boundary assembly and SPD proof
2	Accurate smooth solution?	Four-mesh errors and fitted orders
3	Correct interface physics?	Temperature, normal flux and jump
4	Bounded layered solution?	Extrema and matrix sign pattern
5	Harder linear systems?	Iterations versus mesh and contrast

Sanity checks assess these five tasks using the same smooth and layered benchmarks

Task 1: assembly and matrix structure



$$b_i \leftarrow b_i + \frac{a_{ib} g_b}{h^2}$$

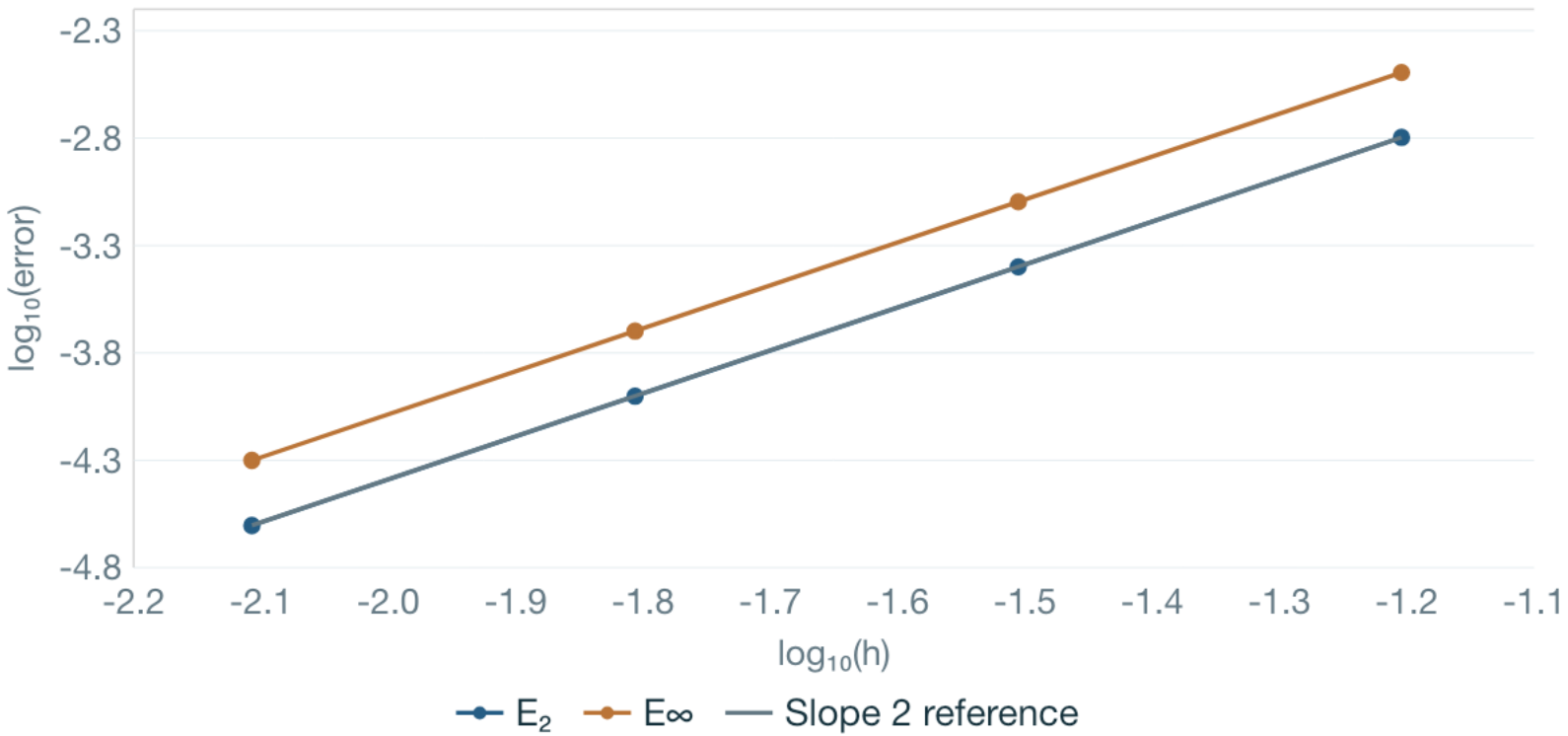
$$h^2 \mathbf{v}^T A_h \mathbf{v} = \sum a_{pr} (v_p - v_r)^2 + \sum a_{pb} v_p^2 > 0$$

$$\frac{\|A - A^T\|_F}{\|A\|_F} = 0$$

N=8: smallest eigenvalue 28.33255
Affine boundary error 6.66e−16

The energy identity proves SPD for nonzero v. The eigenvalue estimate and affine test corroborate the implementation

Task 2: smooth convergence



$$E_2 = h\|U - u_*\|_2, \qquad E_\infty = \max|U - u_*|$$

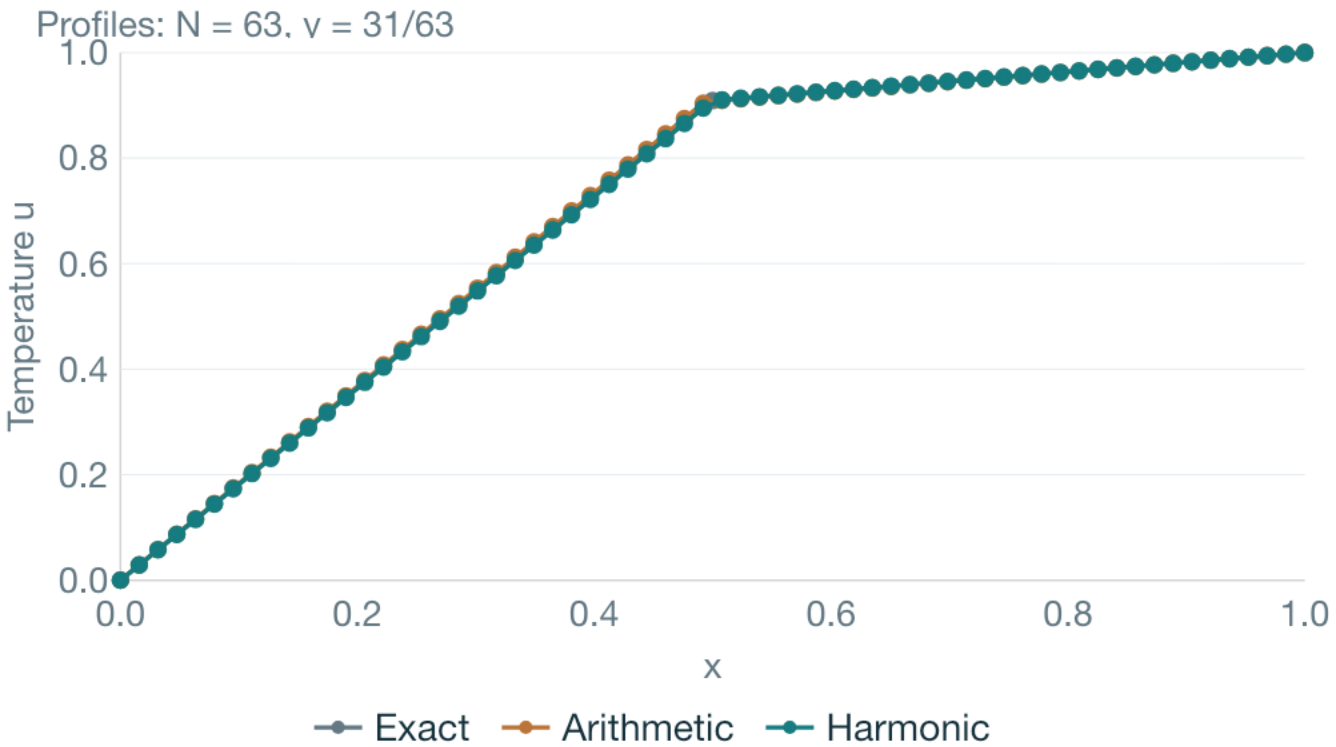
Both axes show base-10 logarithms. Complete error and residual values appear in the backup table

p_2 = 2.00034
$p\infty$ = 1.99984

N = 16, 32, 64, 128

Fit on the last three meshes

Task 3: layered temperature

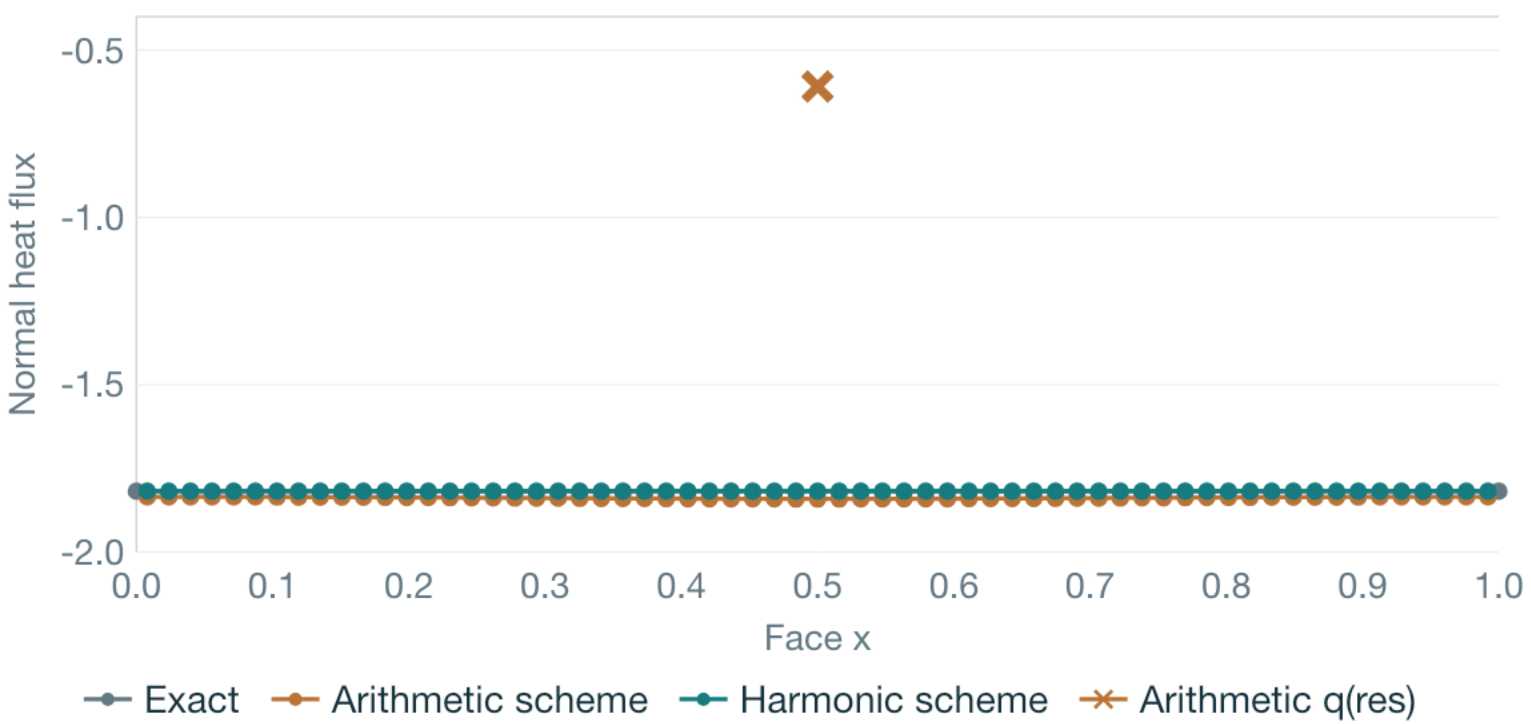


N	Arithmetic E_2	Harmonic E_2
31	6.77e-3	1.32e-15
63	3.27e-3	4.58e-15
127	1.61e-3	2.06e-14
255	7.96e-4	3.88e-14

Arithmetic order ≈ 1.01
Harmonic at round-off level

Same load, full boundary data and LU solver. Both grids remain in [0,1]. Positive shared weights give the maximum principle

Task 3: normal flux and interface jump



N = 63

q(res) = –0.6086
D(phys) = 1.232

$$J = |q_{\text{left}} - q_{\text{right}}|$$

Arithmetic 2.80e-14
Harmonic 4.20e-14

Dphys = max(|qres−qleft|, |qres−qright|). J uses adjacent uncut faces. qres uses the exact series resistance.

Task 4: the maximum principle

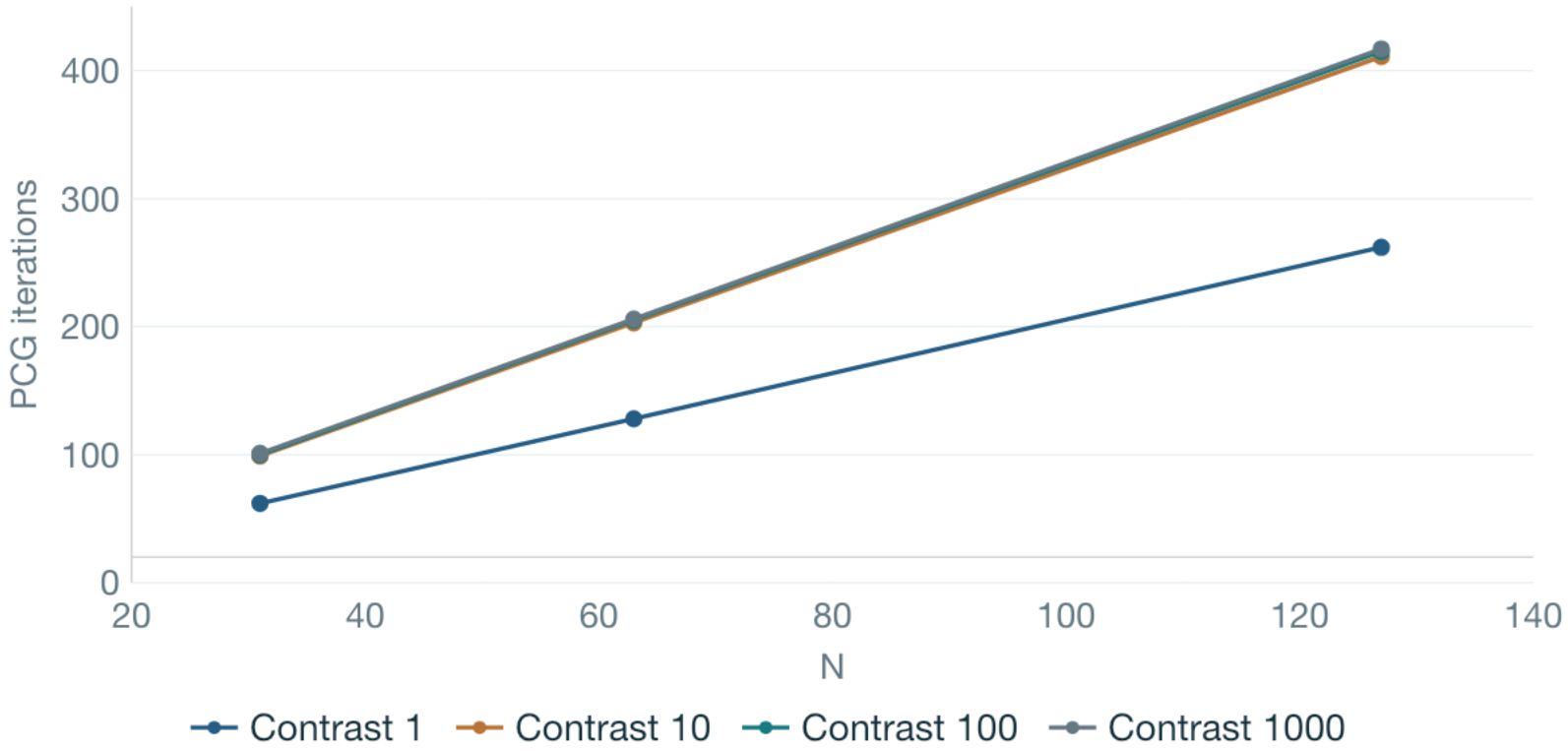
Layered closure	Minimum U	Maximum U
Arithmetic	0	1
Harmonic	0	1

$$A_{ii} > 0, \qquad A_{ij} \leq 0 \quad (i \neq j)$$

M-matrix structure gives the discrete maximum principle

All four meshes, f = 0, boundary range [0,1]. Both closures are bounded, although arithmetic averaging has larger error

Task 5: mesh and coefficient contrast



$f = 1$, $g = 0$
rtol = 10^{-10}
Harmonic faces

Layered geometry
Jacobi scaling
No universal contrast bound

This load differs from the smooth sine benchmark. Iteration counts measure solver work, rather than condition numbers

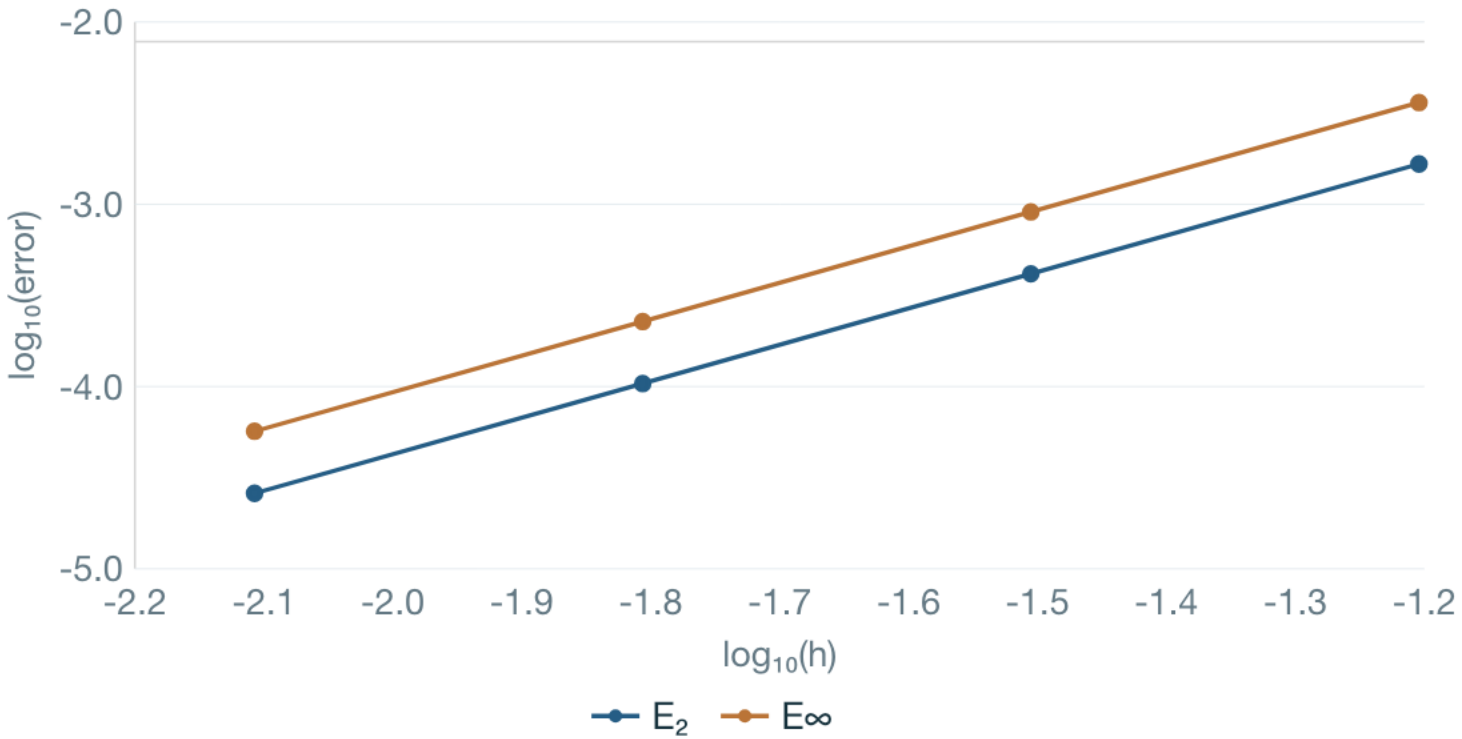
Cost versus accuracy

N	Spatial E_2	LU median (ms)	PCG median (ms)
16	1.60e-3	0.234	0.239
32	3.99e-4	1.281	0.781
64	9.97e-5	6.374	3.162
128	2.49e-5	36.496	19.847

LU slightly faster at N=16. PCG lower median at N≥32

Five samples after one warm-up. LU includes factorisation and solve. PCG includes Jacobi setup. Assembly is excluded

Pure Neumann extension



$$A\mathbf{1} = \mathbf{0}$$

$$\mathbf{1}^T\mathbf{b} = 0$$

$$\text{mean}(U) = 0$$

L_2 order: 2.00035

Zero boundary flux, a=1+x, u*=cos(πx)cos(πy). The code rejects incompatible source b+1 before projection

Sanity checks and conclusions

Expected property	Measured evidence
Smooth convergence near order 2	$p_2 = 2.00034$, $p\infty = 1.99984$
Harmonic nodal reproduction	$E\infty = 3.77e\text{-}15$ to $1.09e\text{-}13$
Nearly constant harmonic flux	Max cut-flux error $\leq 3.73e\text{-}12$
Arithmetic as deliberate comparison	E_2 order ≈ 1.01, physical flux defect persists

Convergence checks accuracy. Resistance checks interface physics.

These checks reuse the two benchmarks. Symmetry and boundedness alone do not establish correct interface physics

A Dirichlet boundary row

$$k = (i-1)(N-1) + (j-1)$$

$$A_{00} = 288, \quad A_{01} = -72, \quad A_{07} = -76$$

$$u = x + 2y, \quad a = 1 + x, \quad f = -1, \quad b_0 = 25$$

N=8, h=1/8. West and south Dirichlet contributions give b_0=25. Unequal x/y entries expose transposed indexing

Timing variability and tolerances

N=128 solver	Min (ms)	Median (ms)	Max (ms)
Sparse LU	36.148	36.496	37.511
Jacobi-PCG	19.574	19.847	20.125

PCG rtol = 10^{-12}
Recomputed relative residual = 2.26e−12

Observed sample ranges describe timing variability. They are not confidence intervals

The mean-zero Neumann system

$$AU + z\lambda = b$$

$$z^T U = 0, \qquad z = \frac{1}{\sqrt{n}}$$

z is the vector of ones divided by √n

Check compatibility before removing round-off imbalance

The saddle system is indefinite, so use sparse LU

The cell-centred graph encodes zero boundary flux directly. The constraint fixes the mean temperature

Four-mesh verification table

N	h	E_2	$E\infty$	ρ	η
16	6.25e-2	1.60e-3	3.21e-3	7.97e-15	1.82e-16
32	3.13e-2	3.99e-4	8.01e-4	3.31e-14	1.85e-16
64	1.56e-2	9.97e-5	2.00e-4	1.37e-13	2.72e-16
128	7.81e-3	2.49e-5	5.01e-5	5.52e-13	2.93e-16

$$\rho = \frac{\|b - AU\|_2}{\max(\|b\|_2, 1)}$$

$$\eta = \frac{\|r\|_\infty}{\|A\|_\infty \|U\|_\infty + \|b\|_\infty}$$

Sparse LU reference has no iterative tolerance. Last-three fits: p_2 = 2.000338, $p\infty$ = 1.999842

Stability and error bounds

$$\tau = A_h u_* - f = O(h^2), \qquad e = U - u_*$$

$$A_h \succeq L_h, \qquad \lambda_{\min}(L_h) \geq 8$$

$$A_h e = -\tau \quad \Longrightarrow \quad E_2(e) \leq \frac{h\|\tau\|_2}{8}$$

$$\psi = y(1-y) \quad \Longrightarrow \quad E_\infty(e) \leq \frac{\|\tau\|_\infty}{8}$$

These coefficient-specific bounds use a=1+x and exact Dirichlet data. The material jump requires resistance analysis

Algebraic versus spatial error

PCG rtol	Iterations	Gap to LU, E_2	Error to u*, E_2
1e-4	78	1.37e-7	9.97e-5
1e-8	88	2.34e-12	9.97e-5
1e-12	94	2.27e-15	9.97e-5

The solver gap shrinks while spatial error stays fixed

N=64, zero initial guess, atol=0. This solver cross-check shares the assembled matrix

Independent verification and limits

Risk	Independent evidence
Wrong forcing	Symbolic derivative and analytic reference
Wrong boundary or index map	Affine solution and hand corner row
Wrong interface resistance	Series-resistance flux reconstruction
Hidden Neumann incompatibility	Negative test with b+1

Limits: square geometry, selected loads, one flux cut and one machine

Ten regression tests pass in the archived project. Raw tables and the complete rebuild script accompany the report

Sources and reproducibility

Project 2 Brief and Problem Pack, Direction 3

Report: conservative variable diffusion and Neumann extension

CSV evidence: smooth, interface, tolerance, solver and Neumann

Rebuild: python rebuild_project.py

AI assistance: drafting, implementation and computational checking.
The team reviews the claims and supplies genuine contribution records.